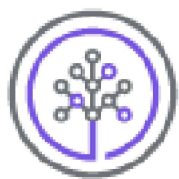


# Cheat Sheet: Plotly and Dash

| Function                             | Description                                                                 | Syntax                                                                                 | Example                                                                                                                            |
|--------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Plotly Express</b>                |                                                                             |                                                                                        |                                                                                                                                    |
| <b>scatter</b>                       | Create a scatter plot                                                       | <code>px.scatter(dataframe, x=x_column, y=y_column)</code>                             | <code>px.scatter(df, x=age_array, y=income_array)</code>                                                                           |
| <b>line</b>                          | Create a line plot                                                          | <code>px.line( x=x_column, y=y_column,'title')</code>                                  | <code>px.line(x=months_array, y=no_bicycle_sold_array)</code>                                                                      |
| <b>bar</b>                           | Create a bar plot                                                           | <code>px.bar( x=x_column, y=y_column,title='title')</code>                             | <code>px.bar( x=grade_array, y=score_array, title='Pass Percentage')</code>                                                        |
| <b>sunburst</b>                      | Create a sunburst plot                                                      | <code>px.sunburst(dataframe, path=[col1,col2..], values='column',title='title')</code> | <code>px.sunburst(data, path=['Month', 'DestStateName'], values='Flights',title='Flight Distribution Hierarchy')</code>            |
| <b>histogram</b>                     | Create a histogram                                                          | <code>px.histogram(x=x,title="title")</code>                                           | <code>px.histogram(x=heights_array,title="Distribution of Heights")</code>                                                         |
| <b>bubble</b>                        | Create a bubble chart                                                       | <code>px.scatter(dataframe, x=x,y=y,size=size,title="title")</code>                    | <code>px.scatter(bub_data, x="City", y="Numberofcrimes", size="Numberofcrimes",hover_name="City", title='Crime Statistics')</code> |
| <b>pie</b>                           | Create a pie chart                                                          | <code>px.pie(values=x,names=y,title="title")</code>                                    | <code>px.pie(values=exp_percent, names=house_holdcategories, title='Household Expenditure')</code>                                 |
| <b>Plotly Graph Objects</b>          |                                                                             |                                                                                        |                                                                                                                                    |
| <b>Scatter</b>                       | Create a scatter                                                            | <code>go.Scatter(x=x, y=y, mode='markers')</code>                                      | <code>go.Scatter(x=age_array, y=income_array, mode='markers')</code>                                                               |
|                                      | Create a line plot                                                          | <code>go.Scatter(x=x, y=y, mode='lines')</code>                                        | <code>go.Bar(x=months_array, y=no_bicycle_sold_array,mode='lines')</code>                                                          |
| <b>add_trace</b>                     | Add additional traces to an existing figure                                 | <code>fig.add_trace(trace_object)</code>                                               | <code>fig.add_trace(go.Scatter(x=months_array, y=no_bicycle_sold_array))</code>                                                    |
| <b>update_layout</b>                 | Update the layout of a figure, such as title, axis labels, and annotations. | <code>fig.update_layout(layout_object)</code>                                          | <code>fig.update_layout(title='Bicycle Sales', xaxis_title='Months', yaxis_title='Number of Bicycles Sold')</code>                 |
| <b>Dash</b>                          |                                                                             |                                                                                        |                                                                                                                                    |
| <b>dash_core_components.Input</b>    | Create an input component                                                   | <code>dcc.Input(value='', type='text')</code>                                          | <code>dcc.Input(value='Hello', type='text')</code>                                                                                 |
| <b>dash_core_components.Graph</b>    | Create a graph component                                                    | <code>dcc.Graph(figure=fig)</code>                                                     | <code>dcc.Graph(figure=fig)</code>                                                                                                 |
| <b>dash_html_components.Div</b>      | Create a div element                                                        | <code>html.Div(children=component_list)</code>                                         | <code>html.Div(children=[html.H1('Hello Dash'), html.P('Welcome to Dash')])</code>                                                 |
| <b>dash_core_components.Dropdown</b> | Create a dropdown component                                                 | <code>dcc.Dropdown(options=options_list, value=default_value)</code>                   | <code>dcc.Dropdown(options=[{'label': 'Option 1', 'value': '1'}, {'label': 'Option 2', 'value': '2'}], value='1')</code>           |

## Author(s)

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# Skills Network